```
}
return 0;
}
```

Output

```
The value of i is 0  The value of j is 10
The value of i is 1  The value of j is 9
The value of i is 2  The value of j is 8
The value of i is 3  The value of j is 7
The value of i is 4  The value of j is 6
```

In the above example, the initialization part contains two variables i and j and update expression contains i++ and j++. These two variables are separated by commas which is executed in sequential order i.e., during initialization firstly i=0 followed by j=10. Similarly, in update expression, firstly i++ is evaluated followed by j++ is evaluated.

Prefer prefix operator over postfix

Generally, the update expression contains increment/decrement operator (++ or --). In this part, always prefer prefix increment/decrement operator over postfix when to be used alone. The reason behind this is that when used alone, prefix operators are executed faster than postfix.

Optional expressions

Generally, the for loop contains three parts, i.e., initialization expressions, test expressions and update expressions. These three expressions are optional in a for loop.

Illustration 10.9 C++ program to sum the numbers from 1 to n

```
#include <iostream>
using namespace std;
int main ()
{
int i, sum=0, n;
cout<<"\n Enter The value of n";
cin>>n;
i =1;
for (  ; i<=n;  )
 {
        sum += i;
        ++i;
 }
cout<<"\n The sum of 1 to " <<n<<"is "<<sum;
return 0;
}
```

Output

```
Enter the value of n 5
The sum of 1 to 5 is 15
```


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